# Introduction: Online Product Ratings and Pricing Analysis for Amazon Electronics Products

- Analyze relationships among price, rating, and review count
- Focus on Amazon electronics-related products
- Explore how customer feedback may reflect product popularity and performance

# Data Sources and Final Analysis Scope

| Source Name | Type | Main Fields | Role in Project |
|---|---|---|---|
| All_Electronics.csv | CSV file | product name, price, rating, review count, category | main analysis |
| Home_Entertainment_Systems.csv | CSV file | product name, price, rating, review count, category | main analysis |
| Amazon Best Sellers webpages | Web scraping | product name, price, rating, review count | supplementary source |
| DummyJSON Products API | API | product name, price, rating, review count, category | API-based source |

For the final quantitative analysis, I mainly used the two structured CSV datasets because they have more consistent schema and pricing format.

# Summery of the results

Number of products: 15,525

Number of sources: 2

Mean price: 2930.07

Median price: 462

Mean rating: 3.99

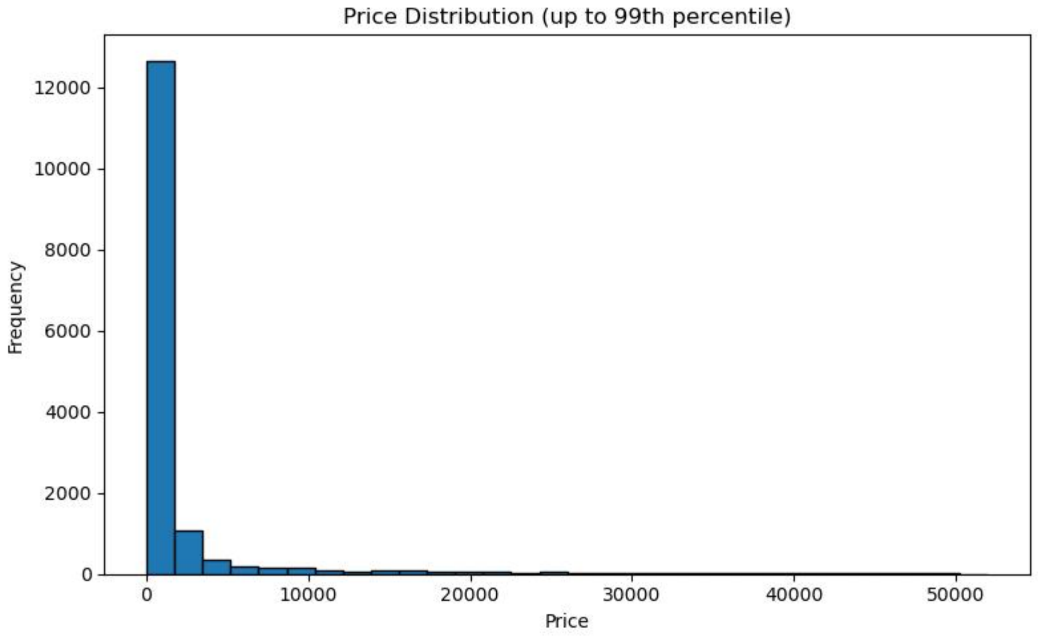
Median rating: 4.10

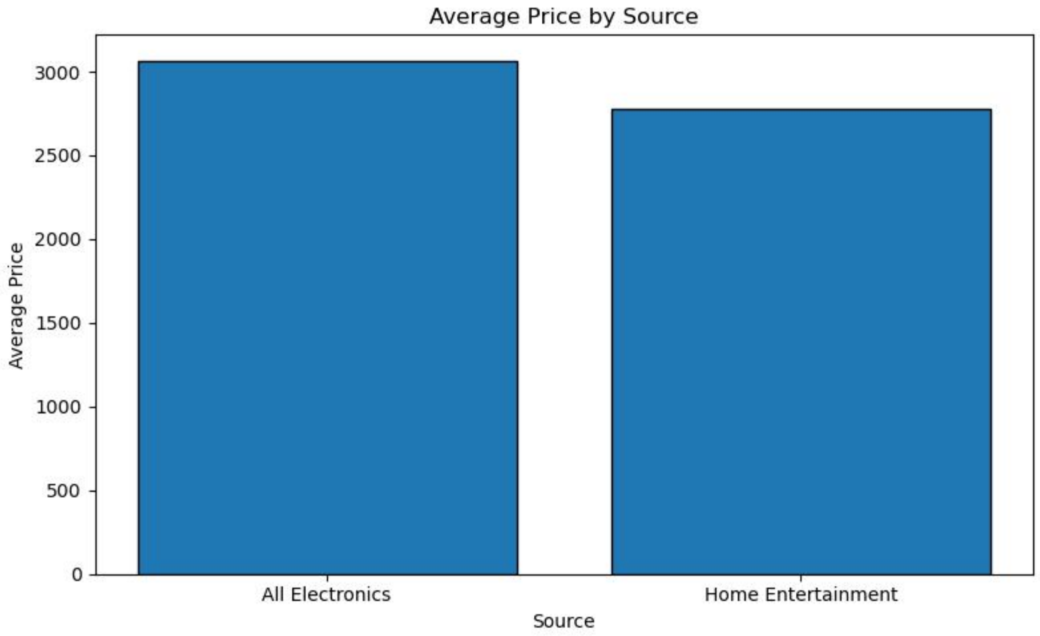
Mean review count: 5569.73

Median review count: 274

- Price is highly right-skewed because the mean is much larger than the median.
- Ratings are more concentrated and remain close to 4.0.

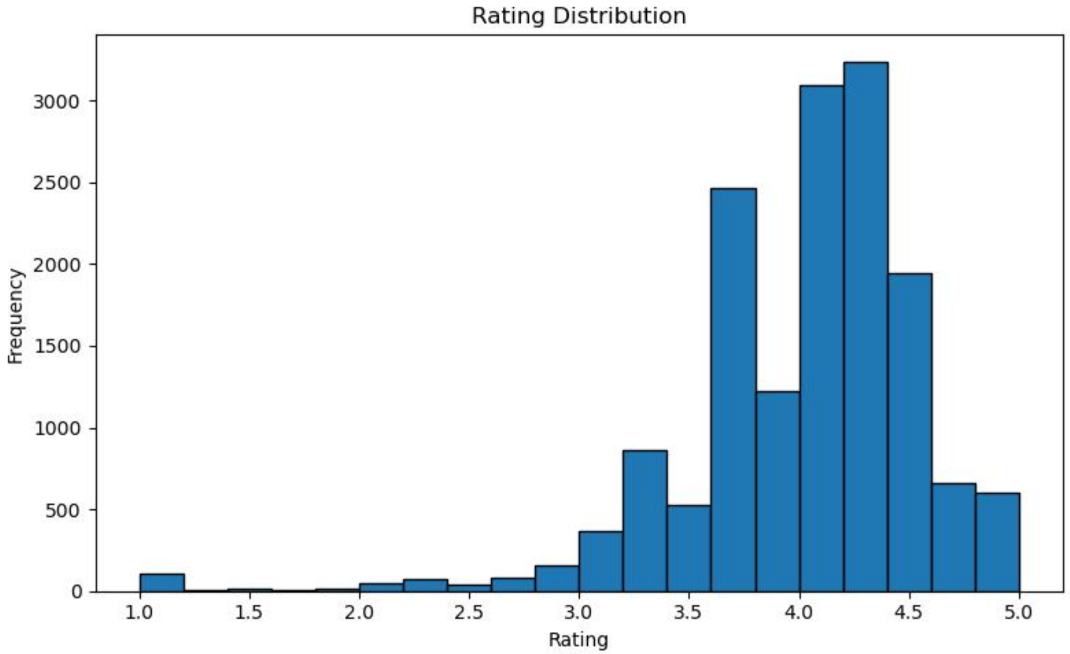
# Price Distribution and Average Price by Source

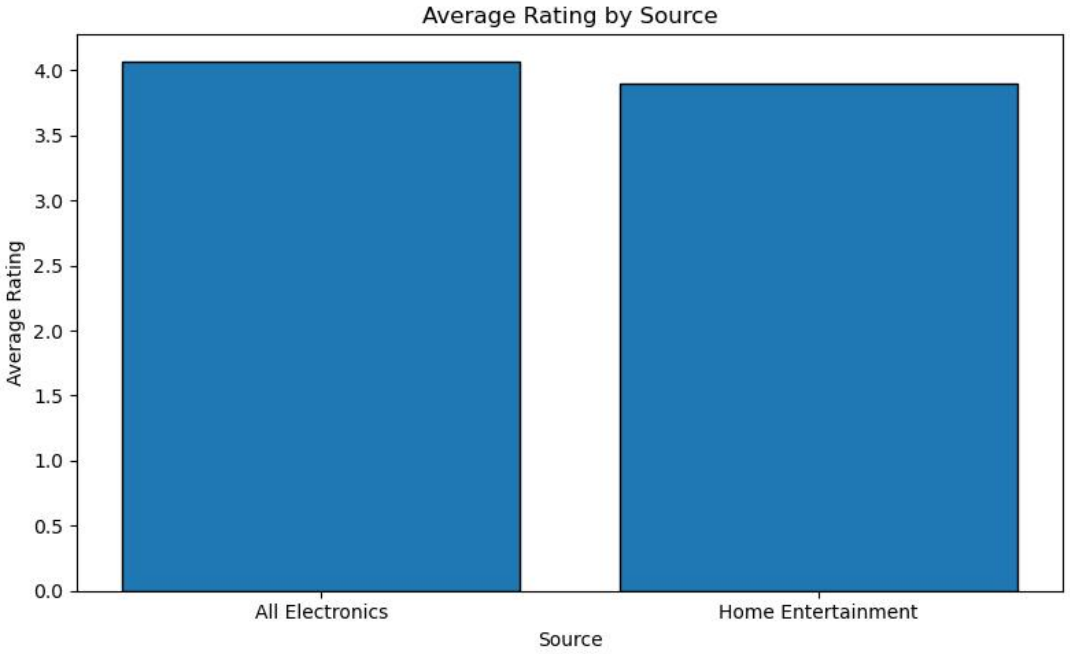



- Most products are concentrated in the low-to-mid price range.
- The price distribution is strongly right-skewed.
- All Electronics has a slightly higher average price than Home Entertainment.

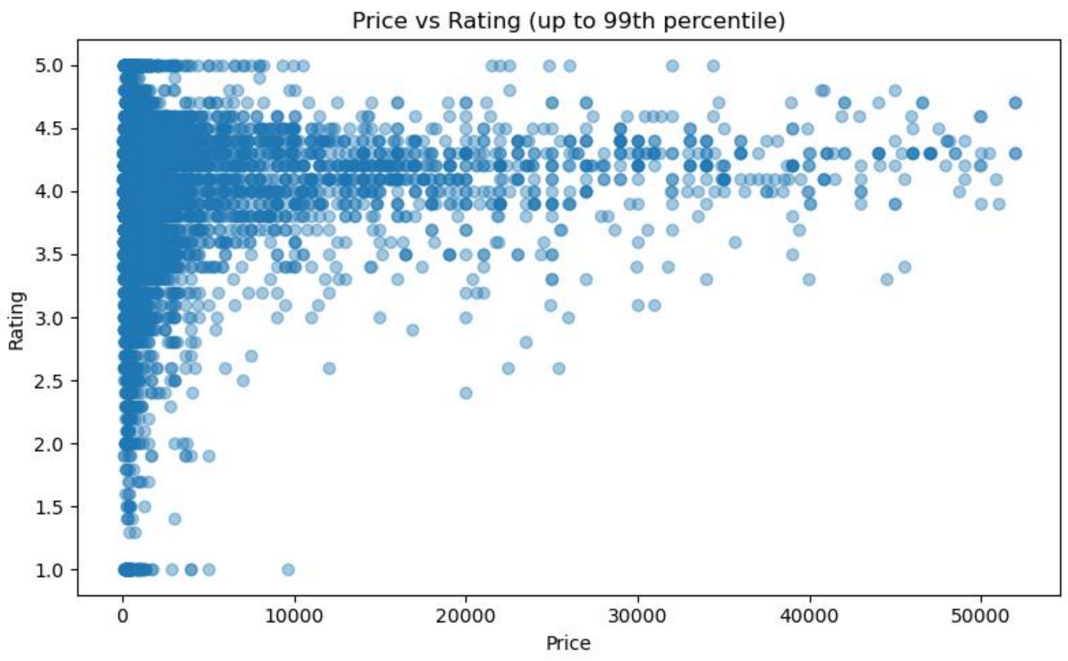
# Rating Distribution and Average Rating by Source

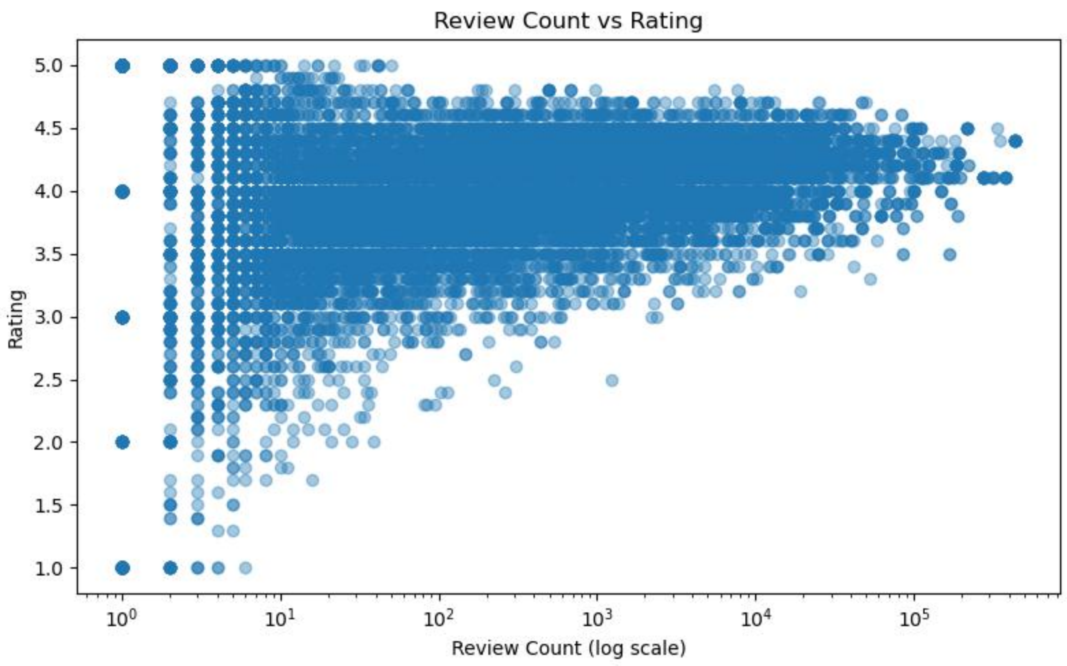



- Most ratings fall between about 4.0 and 4.5.
- Ratings are generally high across products.
- The two sources have very similar average ratings.

# Relationships Among Price, Review Count, and Rating




- Price and rating show only a weak relationship.
- Review count and rating also show a weak overall relationship.
- Products with more reviews tend to have ratings concentrated around 4.0–4.5.

# Challenges

- Different data sources used different column names and formats, so data cleaning and standardization were necessary before analysis.
- Amazon Best Sellers webpage scraping was difficult because webpage structure was inconsistent and each page returned fewer records than expected.
- For the final presentation, I focused the quantitative analysis on the two main CSV datasets because they provided the most consistent structure and pricing format.